Understanding Secondary IP Addressing

You could configure secondary IP addresses to facilitate network migration, support dual subnet operation, and enable seamless transition between legacy and modern IP addressing schemes. This process helps expand the available address space, maintain compatibility with legacy systems, and ensures continuous network connectivity during transitional phases.

A secondary IP address lets a Cisco router interface participate in more than one IP subnet at the same time. This is especially useful for:

- Migrating from an old to a new IP scheme without downtime
- Temporarily expanding available IP addresses on a segment
- Supporting legacy or special gateway requirements

It is ideal for network engineers and IT professionals involved in network migrations, IP address expansion, or temporary workaround implementations. This configuration is especially useful during phased network rollouts, branch office setups, enterprise network restructuring, and lab or simulation environments.

Steps include assigning primary and secondary IP addresses to router interfaces, configuring dynamic routing, verifying IP address assignment, and ensuring proper connectivity and routing between subnets. Proper implementation guarantees optimized network performance, minimal downtime, and streamlined migration processes.

Configuration Syntax Example:

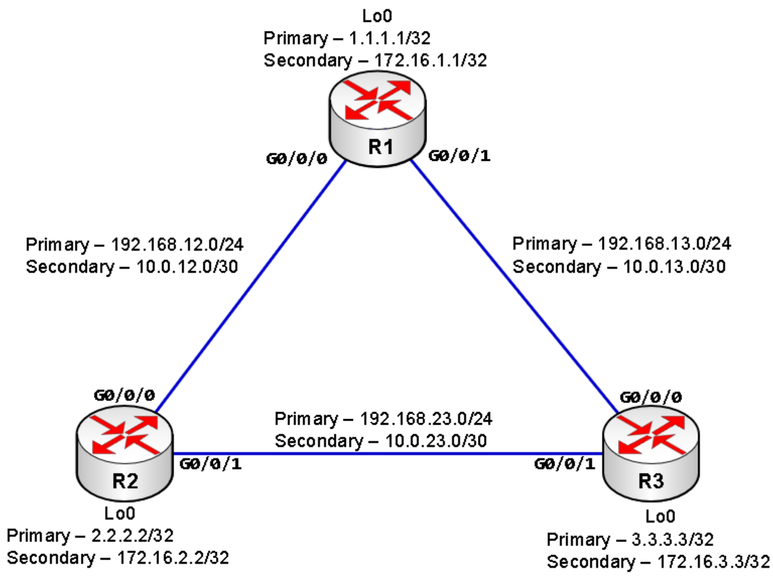
```
Router(config)# interface GigabitEthernet0/0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# ip address 10.10.10.1 255.255.255.0 secondary
Router(config-if)# no shutdown
```

Secondary IP Addressing Lab

Overview

You will use secondary IP addresses to migrate the network to a new subnetting scheme. A secondary IP address lets you assign multiple subnets to a single physical interface. This lab demonstrates how to configure and verify secondary IPs on three Cisco 4331 routers, allowing gradual migration without service disruption.

Topology



Prerequisites

- 3) Cisco 4331 routers (or compatible)
- Console and Ethernet cables
- Terminal software (e.g., PuTTY, HyperTerminal, or compatible)
- Basic knowledge of Cisco IOS and EIGRP

IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
R1	Primary Loopback0	1.1.1.1	255.255.255.255	R1 Loopback/Management
	Secondary Loopback0	172.16.1.1	255.255.255.255	
	Primary GigabitEthernet0/0/0	192.168.12.1	255.255.255.0	R1-R2 Link
	Secondary GigabitEthernet0/0/0	10.0.12.1	255.255.255.252	
	Primary GigabitEthernet0/0/1	192.168.13.1	255.255.255.0	R1-R3 Link
	Secondary GigabitEthernet0/0/1	10.0.13.1	255.255.255.252	
R2	Primary Loopback0	2.2.2.2	255.255.255.255	R2 Loopback/Management
	Secondary Loopback0	172.16.2.2	255.255.255.255	
	Primary GigabitEthernet0/0/0	192.168.12.2	255.255.255.0	R2-R1 L
	Secondary GigabitEthernet0/0/0	10.0.12.2	255.255.255.252	
	Primary GigabitEthernet0/0/1	192.168.23.1	255.255.255.0	R2-R3 Link
	Secondary GigabitEthernet0/0/1	10.0.23.1	255.255.255.252	
R3	Primary Loopback0	3.3.3.3	255.255.255.255	R3 Loopback/Management
	Secondary Loopback0	172.16.3.3	255.255.255.255	
	Primary GigabitEthernet0/0/0	192.168.13.2	255.255.255.0	R3-R1 Link
	Secondary GigabitEthernet0/0/0	10.0.13.2	255.255.255.252	
	Primary GigabitEthernet0/0/1	192.168.23.2	255.255.255.0	R3-R2 Link
	Secondary GigabitEthernet0/0/1	10.0.23.2	255.255.255.252	

Lab Instructions

Step 1: Initial Router Setup

a. Connect via console
b. Configure basic settings:
   - Hostname
   - Disable DNS lookup
   - Passwords and security
   - Enable SSH
     - ip domain-name ccnalab.com
     - username admin password cisco
     - crypto key generate rsa modulus 512
     - ip ssh version 2
   - Console, Auxiliary, and VTY Ports with login local, logging synchronous, exec-timeout, and vty uses SSH only
   - Banner MOTD

Step 2: Configure Primary IP Addresses

a. Assign legacy (192.168.x.x/24) addresses to G0/0/0 and G0/0/1 (see IP Addressing table above).
b. Configure loopback0 with /32 host address.
c. Example:

```
interface GigabitEthernet0/0/0
 description == Link to R2 - Legacy and New Networks ==
 ip address 192.168.12.1 255.255.255.0
 no shutdown
interface Loopback0
 ip address 1.1.1.1 255.255.255.255
```

Step 3: Add Secondary IP Addresses

a. Add new (10.0.x.x/30) addresses as secondary (see IP Addressing table above).
b. Add management subnet to loopback0 with /32 host address as secondary.
c. Example:

```
interface GigabitEthernet0/0/0
 ip address 10.0.12.1 255.255.255.252 secondary
interface Loopback0
 ip address 172.16.1.1 255.255.255.255 secondary
```

Step 4: Configure EIGRP Routing

a. Enable EIGRP (AS 100), advertise all networks (primary and secondary).
b. Example:

```
router eigrp 100
 network 1.1.1.1 0.0.0.0
 network 172.16.1.0 0.0.0.0
 network 192.168.12.0 0.0.0.255
 network 10.0.12.0 0.0.0.3
 passive-interface Loopback0
```

Step 5: Verification and Troubleshooting

a. Use these commands:
  - **`show ip interface brief`**
  - **`show ip interface <interface>`**
  - **`show running-config interface <interface>`**
  - **`show ip route`**
  - **`show ip eigrp neighbors`**
  - **`ping <secondary_ip>`**
  - **`traceroute <destination_ip>`**

Best Practices & Verification

- Use secondary IPs as a temporary solution for migration or expansion, not as a permanent design.
- Ensure all routers on a segment have consistent primary/secondary addressing.
- Routing protocols (EIGRP, OSPF) use the primary address for neighbor relationships.
- Always document and monitor secondary IP usage for troubleshooting and future changes.

Key Use Cases

- IP address expansion: Add more hosts to a segment by introducing a new subnet via secondary IP.
- Migration/re-IP: Enable phased migration from one subnet to another with minimal downtime.
- Merging networks: Allow two subnets to coexist on the same physical interface.
- Legacy/special gateway: Support legacy devices or specific gateway needs.
- Temporary workaround: Address subnet exhaustion or design limitations temporarily.

Lab Completion Checklist

- [ ] All routers configured with hostnames and security.
- [ ] Primary and secondary IPs assigned and verified.
- [ ] EIGRP routing configured for all networks.
- [ ] Ping and traceroute tests successful for all subnets.
- [ ] Routing tables show both legacy and new networks.

Summary

Secondary IP addresses are a powerful tool for network migration and expansion, but should be used with care and as a temporary measure. This lab provides a complete, hands-on guide for configuring and verifying secondary IPs on Cisco 4331 routers in a realistic migration scenario.

Complete Raw Configurations

Router R1

```
enable
config t
hostname R1
enable secret class
banner motd $
****************************************************
UNAUTHORIZED ACCESS IS PROHIBITED
****************************************************
$
no ip domain-lookup
service password-encryption
ip domain-name ccnalab.com
username admin password cisco
crypto key generate rsa modulus 512
ip ssh version 2
line con 0
 login local
 logging synchronous
 exec-timeout 0 0
line aux 0
 login local
 logging synchronous
 exec-timeout 2 0
line vty 0 4
 login local
 transport input telnet ssh
 logging synchronous
 exec-timeout 10 0
 exit
interface Loopback0
 description == R1 Management and Legacy Loopback ==
 ip address 1.1.1.1 255.255.255.255
 ip address 172.16.1.1 255.255.255.255 secondary
interface GigabitEthernet0/0/0
 description == Link to R2 - Legacy and New Networks ==
 ip address 192.168.12.1 255.255.255.0
 ip address 10.0.12.1 255.255.255.252 secondary
 no shutdown
interface GigabitEthernet0/0/1
 description == Link to R3 - Legacy and New Networks ==
 ip address 192.168.13.1 255.255.255.0
 ip address 10.0.13.1 255.255.255.252 secondary
 no shutdown
 exit
router eigrp 100
 network 1.1.1.1 0.0.0.0
 network 172.16.1.0 0.0.0.0
 network 192.168.12.0 0.0.0.255
 network 10.0.12.0 0.0.0.3
 network 192.168.13.0 0.0.0.255
 network 10.0.13.0 0.0.0.3
 passive-interface Loopback0
end
wr
```

Router R2

```
enable
config t
hostname R2
enable secret class
banner motd $
****************************************************
UNAUTHORIZED ACCESS IS PROHIBITED
****************************************************
$
no ip domain-lookup
service password-encryption
ip domain-name ccnalab.com
username admin password cisco
crypto key generate rsa modulus 1024
ip ssh version 2
line con 0
 login local
 logging synchronous
 exec-timeout 0 0
line aux 0
 login local
 logging synchronous
 exec-timeout 2 0
line vty 0 4
 login local
 transport input telnet ssh
 logging synchronous
 exec-timeout 10 0
 exit
interface Loopback0
 description == R2 Management and Legacy Loopback ==
 ip address 2.2.2.2 255.255.255.255
 ip address 172.16.2.2 255.255.255.255 secondary
interface GigabitEthernet0/0/0
 description == Link to R1 - Legacy and New Networks ==
 ip address 192.168.12.2 255.255.255.0
 ip address 10.0.12.2 255.255.255.252 secondary
 no shutdown
interface GigabitEthernet0/0/1
 description == Link to R3 - Legacy and New Networks ==
 ip address 192.168.23.1 255.255.255.0
 ip address 10.0.23.1 255.255.255.252 secondary
 no shutdown
 exit
router eigrp 100
 network 2.2.2.2 0.0.0.0
 network 172.16.2.0 0.0.0.0
 network 192.168.12.0 0.0.0.255
 network 10.0.12.0 0.0.0.3
 network 192.168.23.0 0.0.0.255
 network 10.0.23.0 0.0.0.3
 passive-interface Loopback0
end
wr
```

Router R3

```
enable
config t
hostname R3
enable secret class
banner motd $
****************************************************
UNAUTHORIZED ACCESS IS PROHIBITED
****************************************************
$
no ip domain-lookup
service password-encryption
ip domain-name ccnalab.com
username admin password cisco
crypto key generate rsa modulus 1024
ip ssh version 2
line con 0
 login local
 logging synchronous
 exec-timeout 0 0
line aux 0
 login local
 logging synchronous
 exec-timeout 2 0
line vty 0 4
 login local
 transport input telnet ssh
 logging synchronous
 exec-timeout 10 0
 exit
interface Loopback0
 description == R3 Management and Legacy Loopback ==
 ip address 3.3.3.3 255.255.255.255
 ip address 172.16.3.3 255.255.255.255 secondary
interface GigabitEthernet0/0/0
 description == Link to R1 - Legacy and New Networks ==
 ip address 192.168.13.2 255.255.255.0
 ip address 10.0.13.2 255.255.255.252 secondary
 no shutdown
interface GigabitEthernet0/0/1
 description == Link to R2 - Legacy and New Networks ==
 ip address 192.168.23.2 255.255.255.0
 ip address 10.0.23.2 255.255.255.252 secondary
 no shutdown
 exit
router eigrp 100
 network 3.3.3.3 0.0.0.0
 network 172.16.3.0 0.0.0.0
 network 192.168.13.0 0.0.0.255
 network 10.0.13.0 0.0.0.3
 network 192.168.23.0 0.0.0.255
 network 10.0.23.0 0.0.0.3
 passive-interface Loopback0
end
wr
```